Introduction to Machine Learning

Curse of Dimensionality

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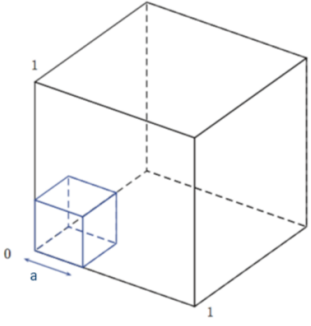
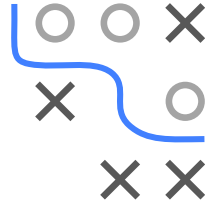


Learning goals

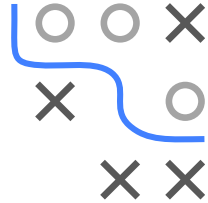
- Understand that our intuition about geometry fails in high-dimensional spaces
- Understand the effects of the curse of dimensionality

CURSE OF DIMENSIONALITY



- The phenomenon of data becoming sparse in high-dimensional spaces is one effect of the **curse of dimensionality**.
- The **curse of dimensionality** refers to various phenomena that arise when analyzing data in high-dimensional spaces that do not occur in low-dimensional spaces.
- Our intuition about the geometry of a space is formed in two and three dimensions.
- We will see: This intuition is often misleading in high-dimensional spaces.

CURSE OF DIMENSIONALITY: EXAMPLE



To illustrate one of the problematic phenomena of data in high dimensional data, we look at an introductory example:

We are given 20 emails, 10 of them are spam and 10 are not. Our goal is to predict if a new incoming mail is spam or not.

For each email, we extract the following features:

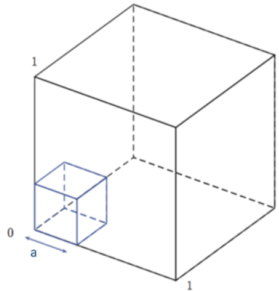
- frequency of exclamation marks (in %)
- the length of the longest sequence of capital letters
- the frequency of certain words, e.g., “free” (in %)
- ...

... and we could extract many more features!

Geometry of High-Dimensional Spaces

THE HIGH-DIMENSIONAL CUBE

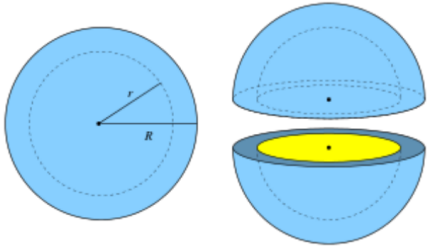
- We embed a small cube with edge length *a* inside a unit cube.
- How long does the edge length *a* of this small hypercube have to be so that the hypercube covers $10\%, 20\%, \ldots$ of the volume of the unit cube (volume 1)?



THE HIGH-DIMENSIONAL SPHERE

Another manifestation of the **curse of dimensionality** is that the majority of data points are close to the outer edges of the sample. Consider a hypersphere of radius 1. The fraction of volume that lies in the ϵ-"edge", $\epsilon := R - r$, of this hypersphere can be calculated by the formula

$$1 - \left(1 - \frac{\epsilon}{R}\right)^p .$$



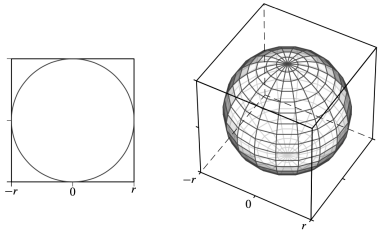
If we peel a high-dimensional orange, there is almost nothing left.

HYPHERSPHERE WITHIN HYPERCUBE

Consider a p-dimensional hypersphere of radius r and volume $S_p(r)$ inscribed in a p-dimensional hypercube with sides of length $2r$ and volume $C_p(r)$. Then it holds that

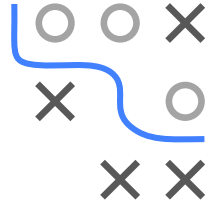
$$\lim_{p\to\infty} \frac{S_p(r)}{C_p(r)} = \lim_{p\to\infty} \frac{\left(\frac{\pi^{\frac{p}{2}}}{\Gamma(\frac{p}{2}+1)}\right) r^p}{(2r)^p} = \lim_{p\to\infty} \frac{\pi^{\frac{p}{2}}}{2^p\Gamma(\frac{p}{2}+1)} = 0,$$

i.e., as the dimensionality increases, most of the volume of the hypercube can be found in its corners.



Mohammed J. Zaki, Wagner Meira, Jr., Data Mining and Analysis: Fundamental Concepts and Algorithms, 2014

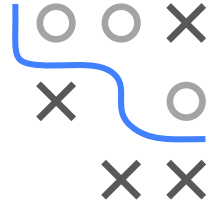
UNIFORMLY DISTRIBUTED DATA



The consequences of the previous results for uniformly distributed data in the high-dimensional hypercube are:

- Most of the data points will lie on the boundary of the space.
- The points will be mainly scattered on the large number of corners of the hypercube, which themselves will become very long spikes.
- Hence the higher the dimensionality, the more similar the minimum and maximum distances between points will become.
- This degrades the effectiveness of most distance functions.
- Neighborhoods of points will not be local anymore.

GAUSSIANS IN HIGH DIMENSIONS



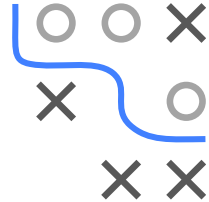
A further manifestation of the **curse of dimensionality** appears if we consider a standard Gaussian $N_p(\mathbf{0}, \boldsymbol{I}_p)$ in p dimensions.

- After transforming from Cartesian to polar coordinates and integrating out the directional variables, we obtain an expression for the density $p(r)$ as a function of the radius r (i.e., the point's distance from the origin), s.t.

$$p(r) = \frac{S_p r^{p-1}}{(2\pi\sigma^2)^{p/2}} \exp\left(-\frac{r^2}{2\sigma^2}\right),$$

  where S_p is the surface area of the p-dimensional unit hypersphere.
- Thus $p(r)\delta r$ is the approximate probability mass inside a thin shell of thickness δr located at radius r.

INTERMEDIATE REMARKS



However, we can find effective techniques applicable to high-dimensional spaces if we exploit these properties of real data:

- Often the data is restricted to a manifold of a lower dimension. (Or at least the directions in the feature space over which significant changes in the target variables occur may be confined.)
- At least locally small changes in the input variables usually will result in small changes in the target variables.